Pulse Transit Time Measurement via Impedance Plethysmography: A Physical Interpretation

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Motivation and Reframing of PTT

A common but imprecise definition of Pulse Transit Time (PTT) describes it as *"the time taken for a pulse to travel from one point to another."* While physically suggestive, this framing leads to ambiguity: *which* two points? *Which* feature of the waveform marks arrival?

A more rigorous and experimentally grounded definition is:

PTT is the time interval between the *difference* of two simultaneously measured Pulse Arrival Times (PATs)—one proximal and one distal with respect to the heart:

$$\mathrm{PTT} = \mathrm{PAT}_{\mathrm{distal}} - \mathrm{PAT}_{\mathrm{proximal}} \tag{1}$$

where PAT_x is the time from the ECG R-peak to the foot of the Arterial Pulse Waveform (APW) at site x.

This formulation avoids reliance on a localized "travel between two points" narrative. Proximal arterial pulse waveform (APW)—i.e., the pulse waveform cor- responding to arterial segments located close to the heart and distal means far away from heart.

Background: Impedance Plethysmography

Impedance Plethysmography (IPG) measures the time-varying electrical impedance of a body segment. A small sinusoidal AC current I is injected through outer electrodes; inner electrodes sense the voltage $V(t)$, from which impedance is derived:

$$Z(t) = \frac{V(t)}{I} = Z_{\mathrm{static}} + \Delta Z_{\mathrm{blood}}(t) \tag{2}$$

Z_{static} arises from tissue, bone, and fat—essentially constant over cardiac timescales. $\Delta Z_{\mathrm{blood}}(t)$ is the small pulsatile component driven by changes in arterial blood volume.

Key physical relationship. Blood is more electrically conductive than surrounding tissue ($\sigma_{\mathrm{blood}} \approx 0.67\,\mathrm{S/m}$ vs. ≈ 0.04–$0.3\,\mathrm{S/m}$ for muscle/fat). When the arterial pressure

wave arrives, the artery expands:

$$\text{arterial volume} \uparrow \Rightarrow \text{ local conductivity} \uparrow \Rightarrow \ Z(t) \downarrow$$

This volume-driven impedance decrease constitutes the IPG waveform, which is functionally analogous to a photoplethysmography (PPG) signal but based on electrical rather than optical principles.

Proximal PAT: Lower-Chest / Thoracic IPG

Electrode Placement

To measure the *proximal* arterial pulse waveform (APW)—i.e., the pulse waveform corresponding to arterial segments located close to the heart, such as the aorta or proximal large arteries—electrodes are placed on the lower chest / sternum region (see Figure **??**). The injected current therefore traverses the thoracic cavity, passing through:

- The **subclavian** arteries

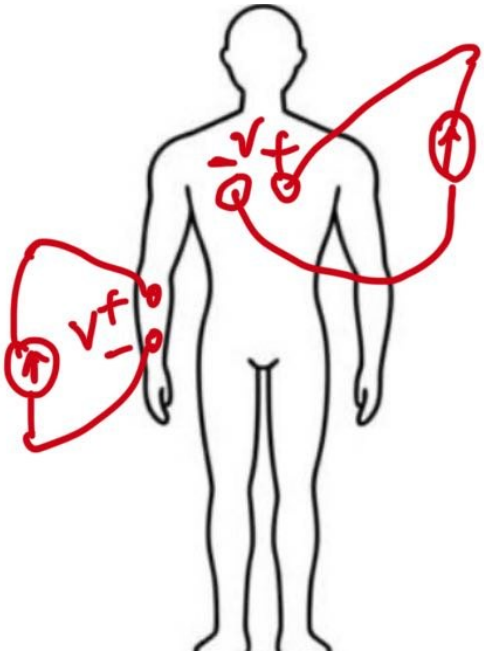


Figure 1: Electrode positioning

How the Impedance Waveform Encodes the APW

During each cardiac cycle, the left ventricle contracts and ejects blood into the aorta. This generates a pressure wave that propagates distally. As the pressure wave arrives at the thoracic aorta:

1. Arterial wall expands ⇒ cross-sectional area of aorta increases ⇒ blood volume in thorax increases.
2. Blood volume ↑ ⇒ local conductivity ↑ ⇒ measured impedance $Z(t)$ *decreases*.
3. This produces the downward deflection of the IPG waveform, mirroring the upstroke of the pressure pulse.

The foot of the APW corresponds to the *onset* of arterial expansion, the very first detectable decrease in impedance, which marks the arrival of the pressure wavefront at the thoracic vessels. This onset time, measured from the ECG R-peak, gives $\text{PAT}_{\text{proximal}}$.

Distal PAT: Forearm or Ankle IPG

Electrode Placement and Rationale

For the *distal* APW, electrodes must be placed such that the current path **does not include the thorax or heart region**. This ensures the measured impedance change is dominated by peripheral, distal vessels. Two common configurations are:

- **Forearm:** Current traverses the brachial artery, radial artery, and ulnar artery.
- **Ankle / lower limb:** Current traverses the posterior tibial artery and dorsalis pedis artery.

Physical Interpretation

The pressure wave, having originated at the heart, travels through the arterial tree and arrives at the peripheral site with a delay. This delay is the propagation time through the intervening arterial segment. When it arrives:

1. Local arterial expansion ⇒ $\Delta Z_{\text{distal}}(t)$ detected by the limb electrodes.
2. Foot of the distal APW = onset of this expansion.
3. $\text{PAT}_{\text{distal}}$ = ECG R-peak → this onset.

Since the current path is confined to the limb, the measured signal is dominated by the relevant peripheral artery (e.g., posterior tibial, dorsalis pedis at ankle), not by central vessels.

Computing PTT and Resolving the "Point-to-Point" Ambiguity

Combining both measurements according to Eq. (1):

$$\boxed{\mathrm{PTT} = \underbrace{\mathrm{PAT}_{\mathrm{distal}}}_{\text{ankle/forearm IPG}} - \underbrace{\mathrm{PAT}_{\mathrm{proximal}}}_{\text{chest IPG}}}$$

This resolves the classical ambiguity as follows:

- We are **not** measuring a pulse travelling between two electrode positions. The electrodes simply define which region of the body the current samples.
- We are measuring **two independently timed volume-change events**, both referenced to the same ECG R-peak.
- The difference of their onset times equals the propagation delay of the pressure wave through the arterial segment between those two vascular regions.

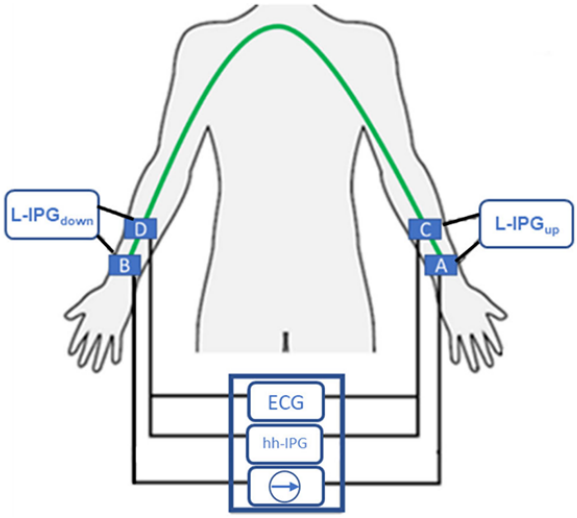


Figure 2: elecrodes only on forearm

Hand-to-Hand IPG: A Special Case for Proximal PAT

An elegant variant is hand-to-hand IPG (hh-IPG), where current is injected between hands. Although the electrodes are peripheral (hands), the current path necessarily passes through the thorax (Fig. **??**), and the signal is dominated by central arterial volume changes—effectively behaving as a *proximal* measurement. This can replace carotid-based PAT (which requires neck sensors) with a more practical wrist-wrist configuration.

The effective impedance can be written as:

$$Z_{\mathrm{hh}}(t) \approx w_{\mathrm{aorta}} Z_{\mathrm{aorta}}(t) + w_{\mathrm{arms}} Z_{\mathrm{arms}}(t) + w_{\mathrm{hands}} Z_{\mathrm{hands}}(t)$$

with $w_{\mathrm{aorta}} \gg w_{\mathrm{arms}}, w_{\mathrm{hands}}$, confirming that hh-IPG reflects proximal hemodynamics despite distal electrode placement.

Summary

Measurement	Electrode site	Dominant artery	PAT type
Chest / hh-IPG	Sternum / hands	Aorta, subclavian	Proximal
Forearm IPG	Forearm cuff	Brachial, radial, ulnar	Distal
Ankle IPG	Ankle cuff	Post. tibial, dors. pedis	Distal

While a pressure pulse *does* physically propagate from heart to periphery, IPG does not measure point-arrival events, it measures a spatially weighted impedance change dominated by specific vascular regions. Electrode geometry determines *which* vascular region is sensed.

On the Necessity of ECG for Foot Detection in PTT Estimation

Pulse Transit Time (PTT) can, in principle, be computed as the time difference between the *foot points* of proximal and distal arterial pulse waveforms (APWs), suggesting that an electrocardiogram (ECG) reference is not strictly required. However, in practice, accurate foot detection is challenging due to noise, baseline drift, and waveform variability. In this work, the ECG R-peak is used to define a temporal search window for locating the APW foot, leveraging the fact that ventricular depolarization precedes pulse wave arrival and thus provides a reliable timing reference. While derivative-based methods can estimate the foot without ECG, they are more sensitive to noise and often yield inconsistent results. Therefore, although ECG is not theoretically necessary for PTT estimation, its use significantly improves the robustness and accuracy of foot detection in practical implementations.

Key References

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3. Mukkamala et al., *Toward ubiquitous blood pressure monitoring via pulse transit time*, IEEE Trans. Biomed. Eng., 62(8), 2015.